

Depth-Damage Curve Model

Depth-Damage, Velocity, Attenuation and Spatial Functions of the Flood Risk Suite

MKM Research Labs

Version 2.0 — 03-November-2026

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Contents

1 Executive Summary **4**

2 Model Purpose and Scope **4**
2.1 Purpose . . . 4
2.2 Scope . . . 4

3 Mathematical Framework **5**
3.1 Depth-Damage Curve . . . 5
3.2 Flood Threshold . . . 5
3.3 Velocity, Retention and the Property Hydrograph . . . 6
3.3.1 Manning's Equation . . . 6
3.3.2 Ground Slope . . . 6
3.3.3 Travel Time . . . 6
3.3.4 Retention with Distance . . . 6
3.3.5 Property Hydrograph . . . 6
3.4 Spatial Functions . . . 7
3.4.1 Haversine Distance . . . 7
3.4.2 Inverse-Distance Weighting . . . 7
3.4.3 Correlation Matrix . . . 7
3.5 Portfolio Analytics . . . 7

4 Input Parameters and Calibration **8**
4.1 Model Parameters . . . 8
4.2 Depth-Damage Calibration . . . 8
4.3 Manning's Roughness . . . 8
4.4 Retention Length . . . 8

5 Implementation Details **9**
5.1 Module Architecture . . . 9
5.2 How Production Uses It . . . 9
5.3 Numerical Considerations . . . 9

6 Sensitivity Analysis **9**
6.1 Depth-Damage Curve Shift . . . 10
6.2 Manning's Roughness . . . 11
6.3 Retention Length . . . 11
6.4 Portfolio Sensitivities . . . 11

7 Validation and Backtesting **11**
7.1 Depth-Damage Curve Validation . . . 11
7.2 Velocity Validation . . . 12
7.3 Implementation Benchmark . . . 12

8 Model Limitations and Known Weaknesses **12**

9 Governance Validation Questions **12**

9.1 Q1 — Ronseal Test . . . 13
9.2 Q2 — Accuracy and Stability . . . 13
9.3 Q3 — Resource Usage . . . 13
9.4 Q4 — Override Controls . . . 13
9.5 Q5 — Output Consumption . . . 13
9.6 Q6 — Model Chaining . . . 13
9.7 Q7 — Weaknesses . . . 13
9.8 Q8 — Monitoring . . . 13
9.9 Q9 — Overall Risk . . . 13

10 Change History **14**

11 Document History **14**

1 Executive Summary

This document presents the mathematical framework, calibration and validation of the Depth-Damage Curve Model (MKM-DD-001), the functions in `src/models/floodrisk/` that turn a flood at a gauge into damage at a property. It documents what the code computes as of package version 2.1, and supersedes v1.1 of this document (filed as *Flood Risk Model*), which still described the property-type factor and the distance-decayed depth calculation removed from the code in May 2026.

The model comprises:

1. **The depth-damage curve** — a UK-calibrated piecewise-linear vulnerability curve from depth above floor level to damage ratio (§3.1).
2. **The flood threshold** — the property's height above its reference gauge plus its floor level (§3.2).
3. **The velocity functions** — Manning's equation, ground slope, travel time, exponential retention with distance, and the single-peak property hydrograph (§3.3).
4. **The spatial functions** — the haversine distance, inverse-distance weighting and an exponential correlation matrix (§3.4).
5. **Portfolio analytics** — a correlated Monte Carlo of portfolio loss with VaR, ES, HHI and clustering (§3.5).

The curve, the threshold and the velocity functions are what production uses: the Property Flood Response model (MKM-PF-001) calls them for every property and storm, and the hazard curve builder applies the curve to gauge levels. The spatial functions other than the haversine, and the portfolio analytics, have no caller in production.

Key sensitivity: the damage curve scales expected loss one for one. A 10% shift of the curve moves a property's expected loss by exactly 10% while its shocked damage ratio cannot reach 1, which holds for damage ratios up to about 0.6; above that the clip at 1 absorbs part of the shift (Table 7). The portfolio figure therefore depends on the portfolio's mix of damage ratios.

2 Model Purpose and Scope

2.1 Purpose

The model quantifies the physical impact of fluvial flooding on a property: how deep the water stands above its floor, when it arrives, and what fraction of reinstatement cost it causes. Those damage ratios feed PRS pricing and the property flood timeseries.

2.2 Scope

- **In scope:** the depth-damage curve; the flood threshold; Manning velocity, slope, travel time and retention; the single-peak property hydrograph; the haversine, IDW and correlation functions; the portfolio Monte Carlo and its metrics.
- **Out of scope:** how a storm's gauge response is chosen and propagated to a property, including the bankfull offset and terrain scaling (MKM-PF-001); the compound per-pulse hydrograph (`floodrisk/hydrograph/`, MKM-HG-001); GEV hazard curve fitting; PRS pricing.
- **Upstream:** gauge responses and timeseries; property elevation and floor level.

- **Downstream:** MKM-PF-001 (property flood response); the hazard curve builder (`src/models/hazard/builder.py`, `terrain_grid.py`).

3 Mathematical Framework

3.1 Depth-Damage Curve

The curve maps depth d above the finished floor (metres) to a damage ratio $\delta \in [0, 1]$, the fraction of reinstatement cost incurred, through ten control points calibrated against UK flood damage survey data:

Depth (m)	0	0.05	0.5	1.0	1.5	2.0	3.0	4.0	5.0	6.0
Damage ratio	0	0.05	0.25	0.40	0.50	0.60	0.75	0.85	0.95	1.0

Table 1: UK-calibrated depth-damage control points

Between adjacent control points $d_i \le d < d_{i+1}$,

$$\delta(d) = \delta_i + \frac{d - d_i}{d_{i+1} - d_i}\,(\delta_{i+1} - \delta_i), \tag{1}$$

with $\delta(d) = 0$ for $d \le 0$ and $\delta(d) = 1$ for $d \ge 6\,\text{m}$. `scalar_depth_damage` evaluates it with a plain loop over the control points (`DEPTH_POINTS`, `DAMAGE_POINTS` in `config/damage.py`).

No property-type factor. Earlier versions scaled the curve by 1.2 for commercial and 0.9 for industrial property. The code no longer does; the factors remain in config as `PROPERTY_TYPE_DAMAGE_FACTORS` but nothing reads them.

The degree-5 curve. `bri_depth_damage` evaluates a degree-5 polynomial through the origin, the least-squares fit to Table 1 ($R^2 = 0.999244$, RMSE 0.009368, largest error 0.020388 at 0.05 m, increasing on $[0, 6]$), on the depth less the floor level and a signed BRI stilt. It is implemented for the BRI models and has no caller in production; the curve of record is Eq. (1).

3.2 Flood Threshold

A property floods when water at it rises above its height above the reference gauge plus its floor level (`relative_elevation`):

$$h_{\text{thr}} = \max(0,\ e_{\text{prop}} - e_{\text{gauge}}) + f, \tag{2}$$

with ground elevations e in metres AOD and floor level f in metres. The depth at the property is the water above that threshold, and the damage ratio is $\delta(\max(0,\ \text{water} - h_{\text{thr}}))$.

Removed in v2.0. Versions 1.x derived depth from the flood WSE, the property's elevation and a linear decay over 25 km from a Thames reference point (their Eqs. 5–6), capped at 5 m. The code replaced that with the nearest-gauge propagation of MKM-PF-001 and Eq. (2); `FLOOD_MAX_DISTANCE_M` remains in config, unread.

3.3 Velocity, Retention and the Property Hydrograph

3.3.1 Manning's Equation

For wide, shallow overbank flow the hydraulic radius is approximated by the depth h:

$$v = \frac{1}{n}\, h^{2/3} S^{1/2}, \tag{3}$$

with $v = 0$ when $h \le 0$ or $n \le 0$, and the slope taken as $\max(|S|, 0.001)$.

3.3.2 Ground Slope

$$S = \max\!\left(\frac{|e_{\text{gauge}} - e_{\text{prop}}|}{d},\ 0.001\right), \tag{4}$$

and $S = 0.001$ when $d \le 0$.

3.3.3 Travel Time

$$t = \frac{d}{3600\, v} \text{ hours}, \tag{5}$$

infinite when $v = 0$.

3.3.4 Retention with Distance

The share of the flood signal that reaches a property at distance d from the river:

$$a(d) = e^{-d/\ell}, \qquad \ell = 10{,}000 \text{ m}, \tag{6}$$

with $a = 1$ for $d \le 0$ and $a = 0$ for $\ell \le 0$. At 600 m a property retains 94% of the signal, at 2 km 82%, at 5 km 61%. There is no near-field bypass: v1.4's rule that $a = 1$ within 2 km has been removed. (v1.1 gave $\ell = 2{,}000$ m.)

3.3.5 Property Hydrograph

`build_property_hydrograph` turns the gauge's hourly series into the property's. With gauge base W_b (its minimum), peak W_p at hour t_p, travel time t_r and the property's peak WSE W^*:

- before the flood front arrives, $t < t_r$, the property sits at W_b with no depth;
- on the rising limb, $t \le t_p + t_r$, the gauge is sampled at the nearest hour to $t - t_r$;
- on the recession, at the nearest hour to $t_p + (t - t_p - t_r)/\gamma$, $\gamma = 1.5$;

and the property's level and depth are

$$W(t) = W_b + (W^* - W_b)\,\frac{W_{\text{gauge}}(\cdot) - W_b}{W_p - W_b}, \qquad d(t) = \max\!\big(0,\ W(t) - e_{\text{prop}} - f\big), \tag{7}$$

both reported to the millimetre. Retention is *not* applied here: the caller (MKM-PF-001) applies $a(d)$ to the water above bankfull before computing W^*, so the hydrograph scales by it once. v1.x's Eq. 15 multiplied by a again.

3.4 Spatial Functions

3.4.1 Haversine Distance

$$d = 2R\,\operatorname{atan2}\!\left(\sqrt{a},\ \sqrt{1-a}\right), \quad a = \sin^2\frac{\Delta\varphi}{2} + \cos\varphi_1\cos\varphi_2\sin^2\frac{\Delta\lambda}{2}, \tag{8}$$

with $R = 6{,}371{,}000\,\text{m}$. This is the one spatial function production calls; it is used across the property, gauge and seismic code.

3.4.2 Inverse-Distance Weighting

Given gauges at distances d_i with water surface elevations W_i, $\hat{W} = \sum_i w_i W_i / \sum_i w_i$ with $w_i = d_i^{-p}$, $p = 2$; the first gauge closer than 1 m returns its own W_i. Production replaced IDW with the single nearest gauge in package v2.1; the function is retained.

3.4.3 Correlation Matrix

For properties at (longitude, latitude) in degrees, with Euclidean separation d_{ij}^{deg} converted at 111 km per degree,

$$\rho_{ij} = \rho_0 \exp\!\left(-\frac{111\,d_{ij}^{\text{deg}}}{L/1000}\right), \quad i \neq j, \qquad \rho_{ii} = 1. \tag{9}$$

The degree conversion takes no account of latitude: at 51.5°N it overstates east–west separation by 61%. The matrix is positive definite.

3.5 Portfolio Analytics

`risk_analytics` simulates portfolio loss under correlated shocks. It has no caller in production; it is documented because it is part of the package.

Monte Carlo. With damage ratios δ_j, values V_j and correlation matrix Σ, it draws $M = 1{,}000$ shock vectors $z \sim N(0, \Sigma)$ with NumPy's legacy generator (seed 42), which factorises Σ by singular-value decomposition, and sets

$$L^{(m)} = \sum_j V_j\,\text{clip}(\delta_j(1 + 0.2 z_j^{(m)}),\ 0,\ 1). \tag{10}$$

VaR_α is the α percentile of the draws (NumPy's linear interpolation) and ES_α the mean of the draws at or above it, for $\alpha = 95, 99$.

The expected loss, exactly. Each shocked ratio $X_j = \delta_j(1 + 0.2 z_j)$ is normal with mean δ_j and deviation $0.2\delta_j$, so

$$\mathbb{E}\,\text{clip}(X_j, 0, 1) = \delta_j\left[\Phi(\beta_j) - \Phi(-5)\right] + 0.2\delta_j\left[\varphi(-5) - \varphi(\beta_j)\right] + 1 - \Phi(\beta_j), \quad \beta_j = \frac{1-\delta_j}{0.2\,\delta_j}, \tag{11}$$

and the expected portfolio loss is $\sum_j V_j\,\mathbb{E}\,\text{clip}(X_j, 0, 1)$ whatever the correlation. The Monte Carlo mean estimates it.

Concentration. The Herfindahl–Hirschman index of losses $\ell_j = V_j \delta_j$ is $\sum_j s_j^2$ with $s_j = \ell_j / \sum_k \ell_k$; it is $1/N$ for an equal portfolio. Grid concentration aggregates by 1 km cells ($1^\circ \approx 111$ km), and KMeans ($K = 5$, ten initialisations, seed 42) clusters properties on standardised depth, damage, position and $\ln(1 + V)$.

4 Input Parameters and Calibration

4.1 Model Parameters

Parameter	Symbol	Default	Where
Manning's roughness	n	0.04	`DEFAULT_ROUGHNESS`
Minimum slope	$S_{\min}$	0.001	`MIN_SLOPE`
Retention length	ℓ	10,000 m	`DEFAULT_RETENTION_LENGTH`
Recession factor	γ	1.5	`DEFAULT_RECESSION_FACTOR`
Earth radius	R	6,371,000 m	`haversine_distance`
IDW power, minimum distance	p, $d_{\min}$	2, 1 m	`idw_interpolate_from_gauges`
Correlation length, base	L, ρ_0	caller's	`build_correlation_matrix`
Monte Carlo draws, seed	M	1,000, 42	`simulate_portfolio_impact`
Shock dispersion	—	0.2	`simulate_portfolio_impact`
KMeans clusters, restarts	K	5, 10	`analyze_risk_clusters`

Table 2: Model parameters (`config/models/_flood.py`, `config/damage.py`)

4.2 Depth-Damage Calibration

The control points of Table 1 follow the Multi-Coloured Manual (Penning-Rowsell et al., 2013), chosen to capture minimal damage below 0.05 m (below-floor wetting), rapid escalation from 0.05 to 1.0 m (electrics, ground-floor contents, plaster), and saturation above 3 m as the structure approaches total loss.

4.3 Manning's Roughness

The default $n = 0.04$ represents a suburban floodplain with mixed land cover (Chow, 1959): smooth concrete 0.012, short grass 0.025, urban and suburban 0.035–0.050, dense vegetation 0.070–0.100.

4.4 Retention Length

$\ell = 10{,}000$ m was chosen so that properties within a few kilometres of the river keep most of the flood signal (82% at 2 km), with the terrain scaling of MKM-PF-001 shortening or lengthening the effective distance by land cover. It is an expert judgement, not fitted to observed inundation extents.

5 Implementation Details

5.1 Module Architecture

Module	Responsibility
`depth_damage.py`	`scalar_depth_damage` (Eq. 1); the BRI polynomial, stilt and floor uplift; `is_prs_flood`
`elevation.py`	`relative_elevation` (Eq. 2)
`velocity.py`	Manning, slope, travel time, retention and the property hydrograph (Eqs. 3–7)
`spatial.py`	haversine, IDW, correlation matrix, KDTree, nearest point on a polyline
`risk_analytics.py`	Monte Carlo, VaR, ES, HHI, grid concentration, KMeans
`hydrograph/`	the compound per-pulse hydrograph (MKM-HG-001; out of scope)

Table 3: Modules of `src/models/floodrisk/` (package version 2.1)

5.2 How Production Uses It

For each property and storm, MKM-PF-001 takes the controlling gauge's peak, subtracts bankfull, scales by the retention $a(d)$ of Eq. (6), compares the result with the threshold of Eq. (2), computes the travel time from Eqs. (3)–(5), builds the hydrograph of Eq. (7), and applies the curve of Eq. (1) to the peak depth. The hazard curve builder applies Eq. (1) to the gauge level above the severe threshold.

5.3 Numerical Considerations

- **Slope floor:** $S_{\min} = 0.001$ keeps Manning's velocity finite on flat ground.
- **Zero velocity:** the travel time is infinite; MKM-PF-001 then uses 0.
- **Damage clipping:** shocked ratios are clipped to $[0, 1]$.
- **Shock draws:** the legacy `np.random.multivariate_normal` factorises by SVD, which tolerates a semi-definite matrix.
- **Reproducibility:** the Monte Carlo reseeds NumPy's global generator with 42 on every call.

6 Sensitivity Analysis

Tables 4–6 are generated from the code; each value is rounded and then printed by size (two places from 1 up, four below).

Table 4: Depth-damage function: damage ratio by flood depth.

Depth (m)	Damage ratio
0.0000	0.0000
0.0500	0.0500
0.1000	0.0722
0.2500	0.1389
0.5000	0.2500
0.7500	0.3250
1.00	0.4000
1.50	0.5000
2.00	0.6000
3.00	0.7500
4.00	0.8500
5.00	0.9500
6.00	1.00

Table 5: Manning's velocity (m/s) by roughness coefficient ($S = 0.005$).

Manning's n	v at $d = 0.5$m	v at $d = 1.0$m	v at $d = 2.0$m
0.0200	2.23	3.54	5.61
0.0300	1.49	2.36	3.74
0.0400	1.11	1.77	2.81
0.0500	0.8910	1.41	2.25
0.0600	0.7420	1.18	1.87
0.0800	0.5570	0.8840	1.40

Table 6: Retention factor by distance and characteristic length.

Distance (m)	$\ell = 3000$m	$\ell = 5000$m	$\ell = 10000$m	$\ell = 20000$m
0	1.00	1.00	1.00	1.00
500	0.8465	0.9048	0.9512	0.9753
1000	0.7165	0.8187	0.9048	0.9512
2000	0.5134	0.6703	0.8187	0.9048
5000	0.1889	0.3679	0.6065	0.7788
10000	0.0357	0.1353	0.3679	0.6065

6.1 Depth-Damage Curve Shift

Scaling every damage ratio by $1 \pm 10\%$ and taking the exact expected loss of Eq. (11) gives, for a single property:

Table 7: Expected loss after a ±10% curve shift, relative to the base curve.

Damage ratio δ	+10%	−10%
0.1–0.5	1.1000	0.9000
0.6	1.0997	0.9000
0.7	1.0947	0.9009
0.8	1.0785	0.9074
0.9	1.0559	0.9228

Below $\delta \approx 0.6$ the shocked ratio almost never reaches 1 and the loss moves exactly with the curve. A portfolio's response is the value-weighted mix of these rows; v1.x's "±8%" corresponds to a portfolio dominated by damage ratios near 0.8.

6.2 Manning's Roughness

At depth 1 m and slope 0.003, over a 3 km flow path:

Table 8: Velocity and travel time by roughness ($d = 1$ m, $S = 0.003$, 3 km).

n	Velocity (m/s)	Travel time (h)
0.025	2.19	0.38
0.040	1.37	0.61
0.060	0.9130	0.91
0.100	0.5480	1.52

Travel time scales with n. v1.x printed velocities of 0.64, 0.40, 0.27 and 0.16 m/s for this case, which Eq. (3) does not give.

6.3 Retention Length

At 2 km from the river a property retains 0.37 of the signal with $\ell = 2{,}000$ m, 0.67 with 5,000 m and 0.82 with the default 10,000 m. Halving ℓ squares the retention at a given distance, so the flooded footprint is sensitive to it; that is why ℓ is a governed parameter.

6.4 Portfolio Sensitivities

The correlation length and Monte Carlo sample size sensitivities of v1.x (its Tables 11 and 12) were computed on a portfolio the document did not specify, and are withdrawn. The Monte Carlo mean estimates Eq. (11), so its sampling error can be judged against that exactly for any portfolio.

7 Validation and Backtesting

7.1 Depth-Damage Curve Validation

As reported in v1.x: the curve lies within the Multi-Coloured Manual's 25th–75th percentile band for residential property at every depth; against Environment Agency post-event surveys (summer

2007, winter 2013/14) its mean absolute error is 6% in damage ratio over 340 properties; and 5-fold cross-validation on the MCM data gives an RMSE of 0.07.

7.2 Velocity Validation

As reported in v1.x: against TUFLOW 2D outputs on three Thames floodplain transects, Manning's velocity is within 18% on average, overestimating in complex urban terrain; travel times are within ±30 minutes over 2 km. The overestimate is conservative, bringing the flood front earlier.

7.3 Implementation Benchmark

The Wolfram oracle benchmark (`MKM-DD-001_wolfram_benchmark`) verifies the code against Eqs. (1)–(11) at 50 digits and through an independent Wolfram Engine evaluation, and reproduces every printed figure in Tables 4–8.

8 Model Limitations and Known Weaknesses

1. **Fluvial only.** Surface water, groundwater, coastal and sewer flooding are not modelled.
2. **Static curve.** Damage depends on depth alone, not on duration, velocity, contamination or warning time.
3. **Piecewise-linear curve.** It may understate damage between 0.05 and 0.5 m, where escalation is nonlinear.
4. **Shallow-flow velocity.** $R_h \approx h$ holds only for wide, shallow overbank flow.
5. **Isotropic retention.** Eq. (6) decays with distance alone, ignoring preferential flow paths; ℓ is expert judgement.
6. **Gaussian dependence.** The Monte Carlo's normal shocks are symmetric and may understate joint tail losses; the 0.2 dispersion is uncalibrated.
7. **Degree conversion.** Eq. (9) and the grid concentration use $1^\circ \approx 111$ km in both directions, overstating east–west distance by 61% at 51.5°N.
8. **No flood defences** and **single events only**.
9. **Unused code.** IDW, the KDTree, the correlation matrix, the portfolio analytics and the BRI polynomial have no production caller, and the property-type factors and 25 km decay radius sit unread in config.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Ronseal Test

Yes, for what production uses: the curve, the threshold and the velocity functions turn a flood at a gauge into depth, timing and damage at a property. The portfolio analytics do what they say but are not used.

9.2 Q2 — Accuracy and Stability

The code computes its documented formulae to rounding error (§7.3); the curve is validated against MCM and EA data and the velocity against TUFLOW (§7.1–7.2). The functions are deterministic; the Monte Carlo is seeded.

9.3 Q3 — Resource Usage

The production functions are scalar and cheap. The correlation matrix is $N \times N$: 800 MB for 10,000 properties.

9.4 Q4 — Override Controls

No manual overrides. Parameters are version-controlled constants in `config/`; changes go through code review.

9.5 Q5 — Output Consumption

Damage ratios and flood depths feed the property flood timeseries and PRS pricing through MKM-PF-001, and the hazard curves through the hazard builder.

9.6 Q6 — Model Chaining

Yes: MKM-PF-001 and the hazard curve builder consume it; it consumes gauge responses and property elevations.

9.7 Q7 — Weaknesses

See §8; the most material are the static curve, the expert-judged retention length and fluvial-only coverage.

9.8 Q8 — Monitoring

Annual review of the curve against EA and MCM data; review of ℓ against observed inundation extents when available; re-running the implementation benchmark on any change to `src/models/floodrisk/`.

9.9 Q9 — Overall Risk

Acceptable, conditional on the review of ℓ: the curve is the most material component and is validated, while the retention length directly sets which properties flood and is not yet fitted to data.

10 Change History

- **v2.0:** documents package v2.1: the property-type factor and the distance-decayed depth removed; the flood threshold of Eq. (2); $\ell = 10{,}000$ m with no near-field bypass; retention applied once, by the caller; SVD shocks; the exact expected loss; the curve-shift, roughness and retention tables regenerated from the code; v1.x's correlation and sample-size tables withdrawn; the degree conversion error corrected to 61%.

11 Document History

Date	Version	Description	Author
13-Feb-2026	1.0	Initial documentation	D. Kelly
18-Feb-2026	1.1	Added Section 9: Governance Validation Questions	D. Kelly
26-Feb-2026	1.2	Delta Engine added to inventory; models in scope updated (not filed)	D. Kelly
26-Mar-2026	1.4	Nearest-gauge propagation; retention length 10 km with a 2 km near-field bypass (not filed)	D. Kelly
03-Nov-2026	2.0	Rewritten to the code (package v2.1) after the Wolfram benchmark	David K Kelly